Building quantum codes with transversal CCZ

from projective geometry

Bohan Lu, Duke University

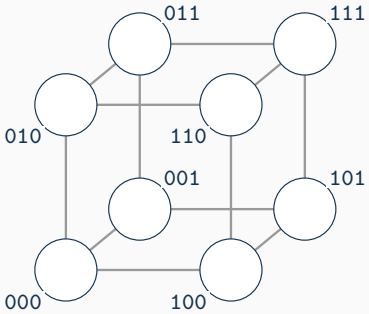
Joint work with Ken Brown

Virginia Tech Applied Algebra Seminar

October 22, 2026

arXiv:2610.09341

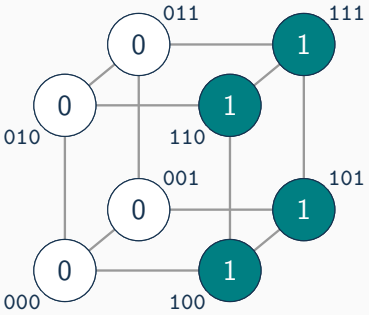
Qubits indexed by the points of $\mathbb{F}_2^3$



Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values at the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$. The overlap $c \wedge c'$ is 1 where both vectors are 1, so it is the vector of the product function cc'.

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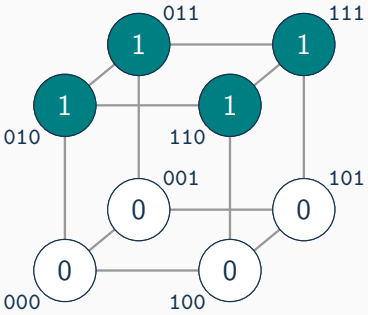
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Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

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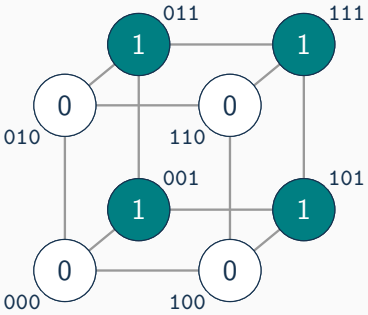
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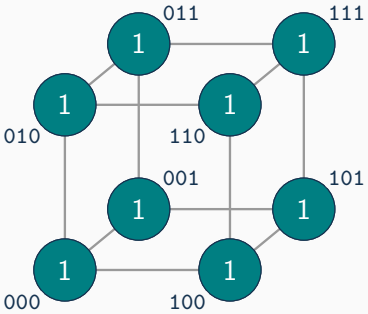
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Qubits indexed by the points of $\mathbb{F}_2^3$



The constant vector $\mathbf{1}$.

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Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

$\mathrm{RM}(r, m)$ is the code of the vectors given by polynomials of degree at most r. For example, $\mathrm{RM}(1, 3) = \mathrm{span}\{\mathbf{1}, u_1, u_2, u_3\}$ is an $[8, 4, 4]$ code.

CSS codes and logical states

Let S $(s \times n)$ and K $(k \times n)$ be binary matrices with $\mathcal{C}_2 = \operatorname{rowspan} S \subset \mathcal{C}_1 = \operatorname{rowspan} \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right]$ and the rows of K independent modulo $\mathcal{C}_2$. The CSS code [CS96, Ste96] has the logical basis

$$|x\rangle_L = |\mathcal{C}_2|^{-1/2} \sum_{v \in \mathcal{C}_2} |xK \oplus v\rangle, \qquad x \in \mathbb{F}_2^k,$$

where $|c\rangle$ is the computational basis state labelled by the binary vector c.

- X_u flips bit u, and $Z_u|c\rangle = (-1)^{c(u)}|c\rangle$. Write $X^v = \prod_u X_u^{v(u)}$ and $Z^z = \prod_u Z_u^{z(u)}$.
- X checks: X^v with $v \in \mathcal{C}_2$. Z checks: Z^z with $z \in \mathcal{C}_1^\perp$.

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
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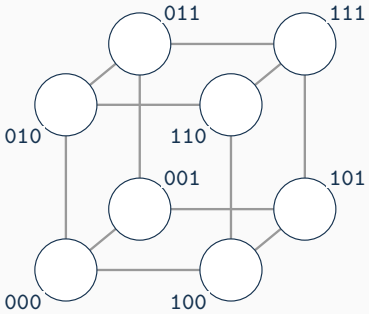
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- X checks: X^v with $v \in \mathcal{C}_2$. Z checks: Z^z with $z \in \mathcal{C}_1^\perp$.
- Logical operators: X^v with $v \in \mathcal{C}_1 \setminus \mathcal{C}_2$, and Z^z with $z \in \mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp$.

$$d_X = \min \operatorname{wt}(\mathcal{C}_1 \setminus \mathcal{C}_2), \qquad d_Z = \min \operatorname{wt}(\mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp), \qquad [\![n, k, d]\!],\ d = \min(d_X, d_Z).$$

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The cube code $[\![8, 3, 2]\!]$

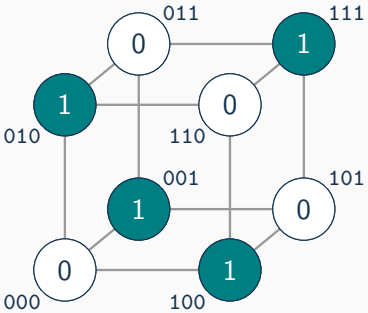


Take $S = \mathbf{1}$, the single check $X^{\otimes 8}$, and let K have the rows u_1, u_2, u_3. Then $\mathcal{C}_1 = \mathrm{RM}(1, 3)$ and $k = 3$.

$xK = x \cdot u = x_1u_1 \oplus x_2u_2 \oplus x_3u_3$, so

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

The cube code $[\![8, 3, 2]\!]$



The codeword $x \cdot u$ for $x = 111$.

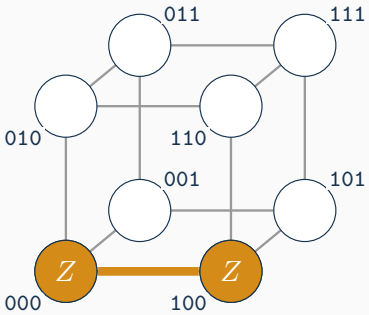
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For $x = 111$ the codeword $x \cdot u$ is $u_1 \oplus u_2 \oplus u_3$.

Distances of the cube code

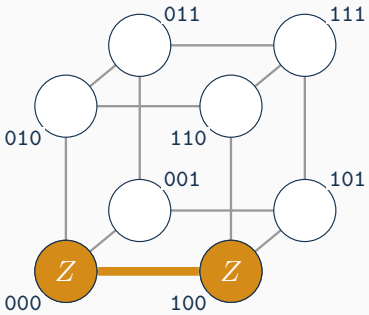


The operator $Z_{000}Z_{100}$.

Let $\mathbf{1}_u$ be the vector with a single 1 at u. Then $Z_{000}Z_{100} = Z^z$ with $z = \mathbf{1}_{000} \oplus \mathbf{1}_{100}$.

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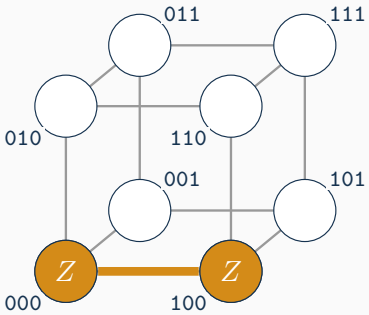
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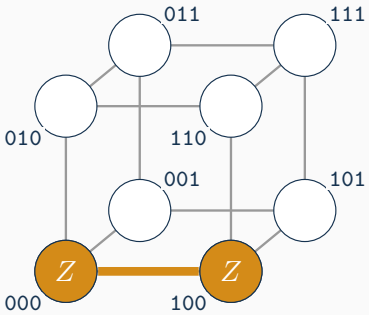
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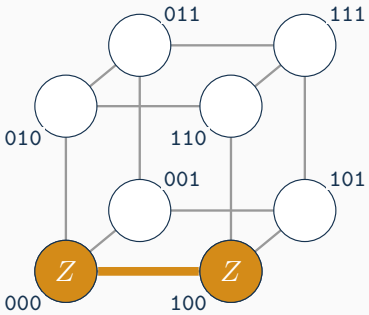
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This is the smallest interesting colour code: a transversal $T/T^\dagger$ layer gives logical CCZ [CH17].

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Transversal gates

A logical gate $\bar{U}$ is *single-qubit transversal* if it is implemented by a product of single-qubit gates,

$$\bar{U} = U_1 \otimes U_2 \otimes \cdots \otimes U_n.$$

A fault in one U_j acts only on qubit j, so it remains a single-qubit error.

The gates in this talk:

$$T = \mathrm{diag}(1, e^{i\pi/4}), \qquad T^\dagger = T^{-1}, \qquad \mathrm{CCZ}|x_1x_2x_3\rangle = (-1)^{x_1x_2x_3}|x_1x_2x_3\rangle.$$

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Question: for which CSS codes with $k = 3$ is there a choice $U_j \in \{T, T^\dagger\}$ such that $U_1 \otimes \cdots \otimes U_n$ acts as logical CCZ?

Codeword weight, physical phase, and logical phase

For $b \in \{0,1\}$, $T^{\pm 1}|b\rangle = e^{\pm i\pi b/4}|b\rangle$. Write $U_j = T^{\Gamma_j}$ with $\Gamma \in \{\pm 1\}^n$. Then

$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi W_\Gamma(c)/4}|c\rangle, \qquad W_\Gamma(c) = \sum_j \Gamma_j c_j.$$

The *signed weight* $W_\Gamma(c)$ counts the ones of c at T sites minus the ones at $T^\dagger$ sites.

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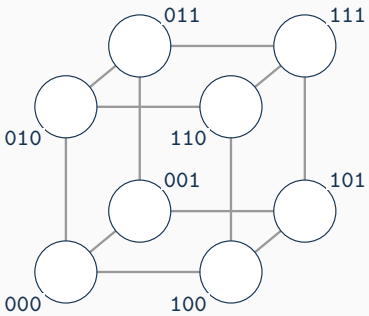
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The product acts as logical CCZ if and only if every codeword of the coset $xK \oplus \mathcal{C}_2$ receives the phase $(-1)^{x_1x_2x_3}$:

$$W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8 \qquad \text{for all } x \in \mathbb{F}_2^3,\ y \in \mathbb{F}_2^s.$$

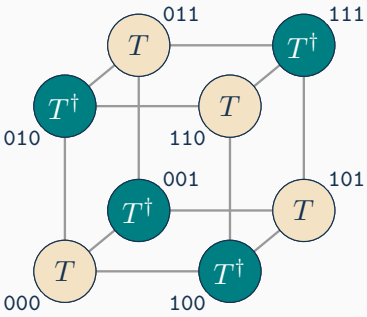
Choosing the signs on the cube



Every codeword of the cube code is an *affine function* $f_{x,b} = b \oplus x \cdot u$. For $x \neq 000$, the hyperplane $x \cdot u = 0$ has 4 of the 8 points, so $f_{x,b}$ has exactly 4 ones.

Let $p = u_1 \oplus u_2 \oplus u_3$. It is the codeword $x \cdot u$ for $x = 111$.

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$T^\dagger$ where $p(u) = 1$ [CH17].

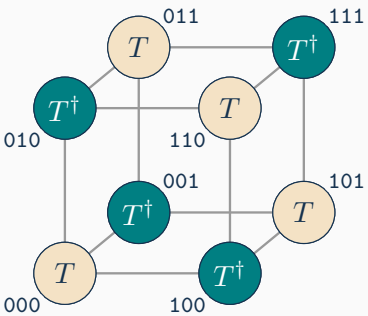
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Put $T^\dagger$ where $p = 1$, so $\Gamma_u = (-1)^{p(u)}$. With the Hamming distance $d_H(p, c) = \mathrm{wt}(p \oplus c)$,

$$W_\Gamma(c) = \mathrm{wt}(c) - 2\,\mathrm{wt}(p \wedge c) = d_H(p, c) - 4.$$

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$x \neq 111$: $p \oplus f_{x,b} = b \oplus (x \oplus 111) \cdot u$ is nonconstant, so $d_H = 4$ and $W_\Gamma = 0$.

$x = 111$: $f_{x,b} \in \{p, \mathbf{1} \oplus p\}$, so $d_H \in \{0, 8\}$ and $W_\Gamma = \pm 4$, phase -1.

Weights of sums expand into overlaps

$$\mathrm{wt}(a_1 \oplus \cdots \oplus a_m) = \sum_{\emptyset \neq J \subseteq \{1,\ldots,m\}} (-2)^{|J|-1}\ \mathrm{wt}\Big(\bigwedge_{j \in J} a_j\Big).$$

Reason: fix one position u and let k of the bits $a_1(u), \ldots, a_m(u)$ be 1. The right side contributes

$$\sum_{t=1}^{k} \binom{k}{t} (-2)^{t-1} = \frac{1-(-1)^k}{2},$$

which is 1 exactly when k is odd, the value of $(a_1 \oplus \cdots \oplus a_m)(u)$. Summing over positions gives the formula. Weighting position u by Γ_u gives the same formula with W_Γ in place of wt.

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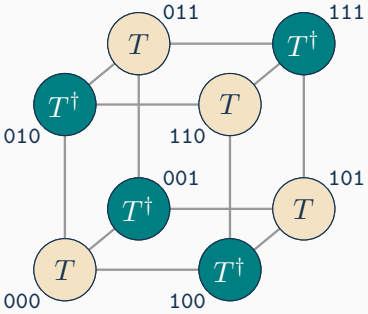
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Since $(-2)^3 = -8$, only the terms with $|J| \leq 3$ remain mod 8.

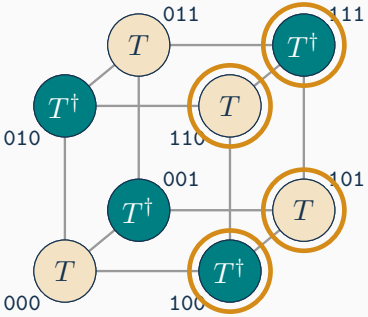
Reading the cube phase from overlaps



Apply the formula to $c = x_1K_1 \oplus x_2K_2 \oplus x_3K_3$:

$$W_\Gamma(c) = \sum_i x_i W_\Gamma(K_i) - 2\sum_{i<j} x_i x_j W_\Gamma(K_i \wedge K_j) + 4x_1x_2x_3\, W_\Gamma(K_1 \wedge K_2 \wedge K_3).$$

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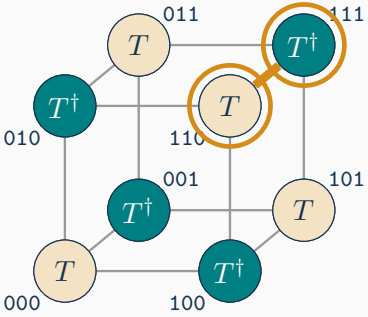
$K_1 = u_1$: the face $u_1 = 1$.

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The face $u_1 = 1$ has T at $101, 110$ and $T^\dagger$ at $100, 111$, so $W_\Gamma(K_1) = 0$. The same holds for K_2, K_3.

Reading the cube phase from overlaps



$K_1 \wedge K_2 = u_1 u_2$.

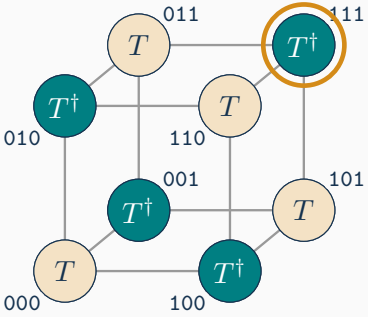
Apply the formula to $c = x_1 K_1 \oplus x_2 K_2 \oplus x_3 K_3$:

$$W_\Gamma(c) = \sum_i x_i W_\Gamma(K_i) - 2 \sum_{i<j} x_i x_j W_\Gamma(K_i \wedge K_j) + 4 x_1 x_2 x_3 \, W_\Gamma(K_1 \wedge K_2 \wedge K_3).$$

The face $u_1 = 1$ has T at $101, 110$ and $T^\dagger$ at $100, 111$, so $W_\Gamma(K_1) = 0$. The same holds for K_2, K_3.

The edge $\{110, 111\}$ has one T and one $T^\dagger$, so $W_\Gamma(K_1 \wedge K_2) = 0$. The same holds for the other pairs.

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The vertex 111 is a $T^\dagger$, so $W_\Gamma(K_1 \wedge K_2 \wedge K_3) = -1$. Hence $W_\Gamma(c) = -4 x_1 x_2 x_3 \equiv 4 x_1 x_2 x_3 \pmod 8$: logical CCZ.

The gate condition becomes conditions on generator rows

Expand $W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8$ with the overlap formula and compare the coefficients of the monomials in (x, y). For distinct rows g_a, g_b, g_c of $G = [K; S]$:

$$W_\Gamma(g_a) \equiv 0 \pmod 8,$$
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The $[\![8, 3, 2]\!]$ code satisfies all three: rows and faces have $W_\Gamma = 0$, edges have $W_\Gamma = 0$, and $u_1u_2u_3$ has weight 1.

Historical context

2005 Bravyi–Kitaev [BK05]: magic-state distillation with ideal Clifford gates.

2012 Bravyi–Haah [BH12]: triorthogonal matrices, transversal T up to Clifford corrections.

2017 Campbell–Howard [CH17]: distillation with multiqubit gate synthesis, including CCZ.

2022 Nezami–Haah [NH22]: classify triorthogonal codes with $n + k \leq 38$ via RM indicators.

2024 Camps-Moreno, López, Matthews et al. [CLM+24]: CSS-T codes from cyclic and extended cyclic codes. In 2026 [CLM+26]: all diagonal transversal gates of a CSS code.

[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

[CLM+24] E. Camps-Moreno et al., An algebraic characterization of binary CSS-T codes and cyclic CSS-T codes for quantum fault tolerance, Quantum Inf. Process. 23, 230 (2024), arXiv:2312.17518.

[CLM+26] E. Camps-Moreno et al., Transversal gates for quantum CSS codes, arXiv:2601.21514 (2026).

Triorthogonal matrix and triorthogonal space

Matrix G [BH12]

$$\mathrm{wt}(g_a \wedge g_b) \equiv 0 \pmod 2,$$
$$\mathrm{wt}(g_a \wedge g_b \wedge g_c) \equiv 0 \pmod 2.$$

Indices are distinct. Odd-weight rows label logical T outputs. Even-weight rows give X checks.

Space $\mathcal{H}$ [NH22]

$$\sum_{j=1}^{L} u_j v_j w_j = 0 \pmod 2$$

for all $u, v, w \in \mathcal{H} \subseteq \mathbb{F}_2^L$, including repeated vectors.

Unital: $\mathbf{1}_L \in \mathcal{H}$. Every vector in $\mathcal{H}$ has even weight.

Nezami–Haah classify the parent spaces $\mathcal{H}$, then obtain codes by puncturing coordinates. This work fixes S from geometry first, then searches for a compatible K.

Conditions for quasi-transversal CCZ

K1	$\mathrm{wt}(K_i) \equiv 0$
K2	$\mathrm{wt}(K_i \wedge K_j) \equiv 0$
K3	$\mathrm{wt}(K_1 \wedge K_2 \wedge K_3) \equiv 1$
KS1	$\mathrm{wt}(K_i \wedge S_l) \equiv 0$
KS2	$\mathrm{wt}(K_i \wedge K_j \wedge S_l) \equiv 0$
KS3	$\mathrm{wt}(K_i \wedge S_l \wedge S_m) \equiv 0$
S1	$\mathrm{wt}(S_l) \equiv 0$
S2	$\mathrm{wt}(S_l \wedge S_m) \equiv 0$
S3	$\mathrm{wt}(S_l \wedge S_m \wedge S_r) \equiv 0$

All mod 2, indices distinct.

Reduce the three row congruences mod 2. Since Γ_j is odd, $W_\Gamma \equiv \mathrm{wt} \pmod 2$, and the result is these nine parities [CH17]. They give CCZ up to a diagonal Clifford correction (*quasi-transversal*).

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A physical $T/T^\dagger$ layer with no correction needs the signed congruences mod 8 and mod 4. These are conditions on Γ as well as on G.

Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ

The construction [LB26]

$Q_{48} = [\![48, 3, 3]\!]$, with $(d_X, d_Z) = (16, 3)$. A layer of $26\,T + 22\,T^\dagger$ implements logical CCZ with no Clifford correction.

- The stabilizer code is fixed by projective geometry: $\mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2)$.
- The logical rows come from a SAT search.
- Lower bound: no code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$. Lengths 39 to 46 are excluded by [WJS26].
- We found codes at block lengths 48 to 496 using S constructed from $\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2)$.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).

Fix S using finite/projective geometry code

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4. Solve for the signs Γ over $\mathbb{Z}_4$.

Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error at site j has syndrome β_j.

$$\beta_j \neq 0 \implies \text{weight-one errors are detected}, \qquad \beta_i \neq \beta_j \implies \beta_i \oplus \beta_j \neq 0.$$

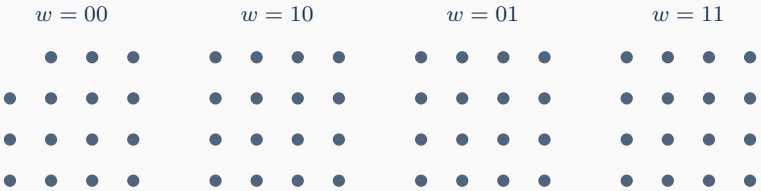
Distinct nonzero columns are points of $\mathrm{PG}(s-1,2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$.

$d_Z \geq 3$ follows from the stabilizer columns, independent of the choice of logical rows.

Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

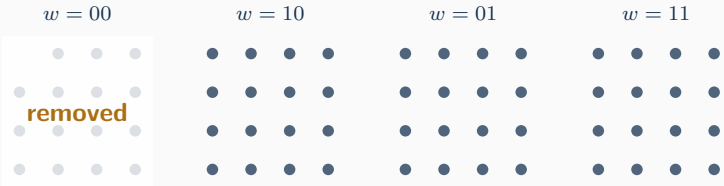
Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$.



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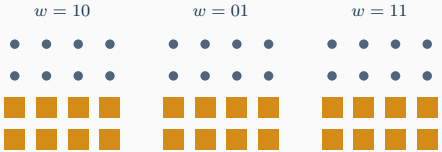


The 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$ are the rows of S.

Three blocks of 16 points give a projective $[48, 6]$ stabilizer code.

Hyperplane sections and codeword weights

A codeword of $\mathcal{C}_2$ evaluates $\ell(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $\mathrm{wt}(\ell) = |\mathcal{P}| - |\mathcal{P} \cap \{\ell = 0\}|$.



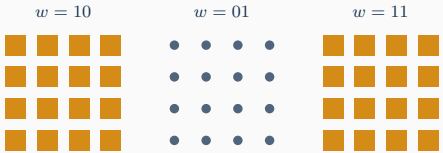
$\lambda \neq 0$, for example $\ell = u_1$: eight points in each block, weight $3 \cdot 8 = 24$.

Projective two-weight codes correspond to finite-geometry sets with two hyperplane intersection sizes [CK86].

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

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$\lambda = 0$, for example $\ell = w_1$: two full blocks, weight $2 \cdot 16 = 32$.

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Weights 24 and 32 are multiples of 8, so S1–S3 hold.

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

More geometric constructions

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q-1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

$p\backslash q$	1	2	3	4	5
4	none	$[\![48,3,3]\!]$, $2^\star$	$[\![112,3,3]\!]$, 12	$[\![240,3,3]\!]$, 17	$[\![496,3,3]\!]$, 18
5	none	$[\![96,3,3]\!]$, 11	$[\![224,3,3]\!]$, 8	$[\![480,3,3]\!]$, 14	
6	$[\![64,3,4]\!]$, 5	$[\![192,3,3]\!]$, 14	$[\![448,3,3]\!]$, 13		
7	$[\![128,3,4]\!]$, 3	$[\![384,3,3]\!]$, 18			
8	$[\![256,3,4]\!]$, 3				

Each cell: parameters with $d = d_Z < d_X$, and the number of inequivalent codes found. Counts are lower bounds except $\star$ (complete). "None": excluded by our lower bound. Supports with $p \leq 3$ admit no valid code.

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. Write χ_M as a multilinear polynomial. Then $\chi_M \in \mathrm{RM}(\deg \chi_M, r)$ and $\mathrm{wt}(\chi_M)$ is the number of columns.

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For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, every x occurs with $y = 1$ and no column has $y = 0$. So $\chi_G(x, y) = 1$ exactly when $y = 1$, that is $\chi_G(x, y) = y$.

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Q_{48}: the columns of S are the points with $w \neq 00$, so

$$\chi_S(u, w) = 1 + (1 + w_1)(1 + w_2) = w_1 + w_2 + w_1 w_2.$$

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g\rangle = \sum_v f(v)g(v) \bmod 2$ [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t\in T} z_t$ and

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Since $\mathrm{RM}(3, r)^\perp = \mathrm{RM}(r-4, r)$, and $y_1 \cdots y_s$ (one at the 8 points $(x, 1, \ldots, 1)$) is one solution [LB26],

$$\chi_G \in y_1 \cdots y_s + \varepsilon\delta_0 + \mathrm{RM}(r-4, r), \qquad r = 3 + s,\ \varepsilon = n \bmod 2.$$

Logical rows in the column picture

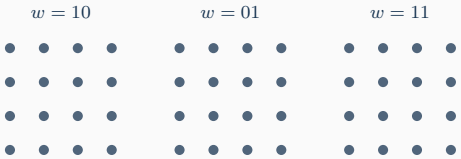
Column $\beta = (u, w) \in \mathcal{P}$ of S carries a *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the column of K above it [LB26]:

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Split a set T of rows into its logical rows T_K and stabilizer rows T_S. With $[P] = 1$ if P holds and 0 otherwise, the CH conditions become

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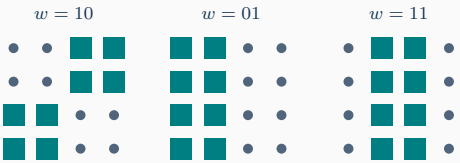
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Teal: $\kappa_1 = 1$, with $\kappa_1 = u_1 \oplus u_3$, $1 \oplus u_3$, u_4 on the three blocks.

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) g(\beta) \bmod 2$ [LB26].

1. K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l \beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2, s))^{\perp}$, a linear condition.

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2. Choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ with $\langle \kappa_1 \kappa_2, \mathbf{1} \rangle_{\mathcal{P}} = \langle \kappa_1 \kappa_2, \beta_l \rangle_{\mathcal{P}} = 0$ (K2 and KS2).

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3. For a fixed pair, the remaining conditions on $\kappa_3 \in V_{\mathcal{P}}$, namely $\langle \kappa_3, \kappa_i\rangle_{\mathcal{P}} = \langle \kappa_3, \beta_l\kappa_i\rangle_{\mathcal{P}} = 0$ and $\langle \kappa_3, \kappa_1\kappa_2\rangle_{\mathcal{P}} = 1$, are linear. Gaussian elimination over $\mathbb{F}_2$ finds κ_3 or shows that none exists.

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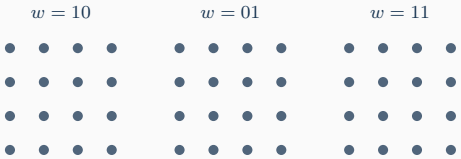
1. K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l\beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2,s))^{\perp}$, a linear condition.
2. Choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ with $\langle \kappa_1\kappa_2, \mathbf{1}\rangle_{\mathcal{P}} = \langle \kappa_1\kappa_2, \beta_l\rangle_{\mathcal{P}} = 0$ (K2 and KS2).
3. For a fixed pair, the remaining conditions on $\kappa_3 \in V_{\mathcal{P}}$, namely $\langle \kappa_3, \kappa_i\rangle_{\mathcal{P}} = \langle \kappa_3, \beta_l\kappa_i\rangle_{\mathcal{P}} = 0$ and $\langle \kappa_3, \kappa_1\kappa_2\rangle_{\mathcal{P}} = 1$, are linear. Gaussian elimination over $\mathbb{F}_2$ finds κ_3 or shows that none exists.

If no pair extends, $\mathcal{P}$ is excluded. A SAT solver can also solve the full system at once. For Q_{48}, the columns at $u = 0000$ in the three blocks sum to 0 and their labels to $011 \oplus 110 \oplus 000 = 101$. These three Z errors form a logical operator, so $d_Z = 3$.

Solving for the $T/T^\dagger$ pattern

Write $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j \in \{0, 1\}$. Then $W_\Gamma(c) = \mathrm{wt}(c) - 2\,\gamma \cdot c$, and every codeword $c = xK \oplus yS$ has even weight:

$$\gamma \cdot c \equiv \tfrac{1}{2}\,\mathrm{wt}(c) - 2x_1x_2x_3 \pmod 4.$$



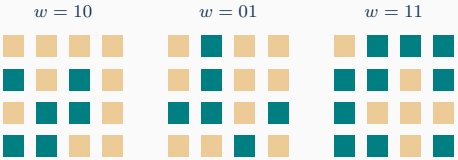
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For Q_{48} [LB26]:

$$\gamma(u, w) = u_3 \oplus w_1 Q_1(u) \oplus w_2 Q_2(u),$$
$$Q_1 = u_1 \oplus u_2 \oplus u_3 \oplus u_1u_3 \oplus u_2u_3 \oplus u_2u_4,$$
$$Q_2 = u_3 \oplus u_4 \oplus u_1u_2 \oplus u_1u_4 \oplus u_3u_4.$$



Teal: $T^\dagger$ $(6, 6, 10)$. Gold: T. Total $26\,T + 22\,T^\dagger$.

Stabilizer supports are Reed–Muller codewords

For projective S with support $\mathcal{P} \subset \mathbb{F}_2^s$, S1–S3 imply

$$\sum_{\beta \in \mathcal{P}} \beta_{i_1} \cdots \beta_{i_t} = 0 \pmod 2, \qquad 1 \le t \le 3.$$

Following [NH22], let $c = \chi_{\mathcal{P}} + \epsilon\delta_0$, where δ_0 is the indicator of the origin and ϵ makes $\mathrm{wt}(c)$ even. Every monomial of degree at least 1 vanishes at $\beta = 0$, so adding δ_0 changes only $\langle c, \mathbf{1} \rangle = \mathrm{wt}(c) \bmod 2$, which ϵ makes 0. Hence c is orthogonal to every monomial of degree at most 3, so $c \perp \mathrm{RM}(3, s)$ and

$$c \in \mathrm{RM}(3, s)^{\perp} = \mathrm{RM}(s - 4, s), \qquad n = \mathrm{wt}(c) - c(0).$$

For Q_{48}: $c = w_1 + w_2 + w_1 w_2 \in \mathrm{RM}(2, 6)$.

Our exclusion argument covers every length through 38

1. Reduce stabilizer rank to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the weight spectrum and support geometry.
3. At rank 8, reduce to finitely many normal forms and enumerate the remaining cases.
4. Show that no remaining support admits the required logical triple.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Existence of a valid code at different block lengths

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$, no correction	Our explicit code

Wills–Jain–Singh classify $n \leq 54$, $d_Z \geq 3$.

The $[\![47, 3, 3]\!]$ code of [JVB+26] has no physical $T/T^\dagger$ layer [LB26]. Its stabilizer support is $\mathbb{F}_2^6$ minus an affine 4-flat, punctured at the origin. Q_{48} removes a linear 4-dimensional subspace instead. [JVB+26] report a 47-to-1 CCZ distillation protocol from it with circuit footprint 9.

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

Takeaways and open questions

Takeaways

- Projective geometry fixes S: $d_Z \geq 3$ and weights divisible by 8.
- Column indicators turn the CH conditions into Boolean equations. The signs solve a $\mathbb{Z}_4$ system.
- $Q_{48} = [\![48, 3, 3]\!]$ with $26\,T + 22\,T^\dagger$. Ours excludes $n \leq 38$, [WJS26] excludes 39–46.

Open questions

- Which supports admit compatible labels and signs?
- Higher distance, other gates, more logical qubits.
- Minimum length for $T/T^\dagger$ CCZ with no correction.
- Full circuits, noise, and resource costs.

Paper
arXiv:2610.09341

Code and data
github.com/jerrylvx/gate2code-public